

Worksheet

1. Does the sequence $\left\{\left(\frac{4}{5}\right)^k\right\}$ converge? If so what is the limit? If not, why not?

Yes $\lim_{k \rightarrow \infty} \left(\frac{4}{5}\right)^k = \lim_{k \rightarrow \infty} \frac{4^k}{5^k} = 0$

2. Does the series $\sum_{k=0}^{\infty} \left(\frac{4}{5}\right)^k$ converge? If so what does it converge to? If not, why not?

Yes. It is geometric. Since $\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}$, ← known formula if $|r| < 1$

$$\sum_{k=0}^{\infty} \left(\frac{4}{5}\right)^k = \frac{1}{1 - \frac{4}{5}} = 5$$

3. Does the series $\sum_{k=2}^{\infty} 3 \left(\frac{4^{k+2}}{5^k}\right)$ converge? If so what does it converge to? If not, why not?

$$\sum_{k=2}^{\infty} 3 \left(\frac{4^{k+2}}{5^k}\right) = \sum_{k=2}^{\infty} 3 \cdot 4^2 \cdot \frac{4^k}{5^k} = \sum_{k=2}^{\infty} 48 \left(\frac{4}{5}\right)^k = \frac{48}{1 - \frac{4}{5}} = 48 - 48\left(\frac{4}{5}\right)$$

4. Does $\sum_{k=1}^{\infty} 5 \frac{3^{k-2}}{2^{k+1}}$ converge? If so what does it converge to? If not, why not?

$$\sum_{k=1}^{\infty} 5 \frac{1}{2 \cdot 3^2} \frac{3^k}{2^k} = \sum_{k=1}^{\infty} \frac{5}{18} \left(\frac{3}{2}\right)^k \quad \text{diverges b/c } \left|\frac{3}{2}\right| > 1$$

5. Does $\sum_{k=1}^{\infty} 5 \frac{2^{k-2}}{7^{k+1}}$ converge? If so what does it converge to? If not, why not?

$$\sum_{k=1}^{\infty} 5 \frac{1}{2^2 \cdot 7} \left(\frac{2}{7}\right)^k = \sum_{k=1}^{\infty} \frac{5}{28} \left(\frac{2}{7}\right)^k = \frac{\frac{5}{28}}{1 - \frac{2}{7}} = \frac{5}{28}$$